INTERNET OF THINGS

The term "Internet of Things" (IoT) was coined by Kevin Ashton at a presentation to Proctor & Gamble in 1999.

The **Internet of Things** is an emerging topic of technical, social and economic importance. Consumer products, durable goods, cars and trucks, industrial components and facilities, sensors, and other everyday objects are combined with internet connectivity and powerful data analysis capabilities that promise to transform the way we live and work.

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INTERNET OF THINGS (IOT)

- Is a network of physical devices that can transfer data to one another **without human intervention**.
- The IOT concept was coined by a member of the Radio Frequency Identification(RFID) development community, the computer scientist Kevin Ashton in 1999.
- IoT devices are not limited to computers or machinery
- IoT can include anything with a sensor that is assigned a unique identifier (UID).
- The main goal of the IoT is to create self-reporting devices that can communicate with each other (and users) in real time.
- **IoT is about data, devices, and connectivity**
- IoT is to provide **Any TIME**, **Any PLACE**, and **Any THING** connections.

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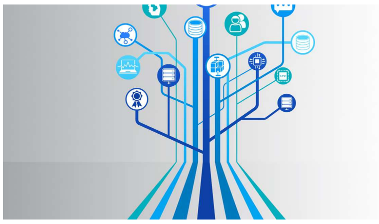
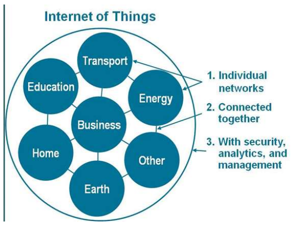
INTERNET OF THINGS (IOT)

- “The Internet of Things (IoT) is the physical objects’ network that takes advantage of the connection of machines and sensors to exchange data in real time with other IoT devices based on the Internet. IoT devices can refer to anything from wearable fitness trackers to smart home devices to industrial equipment in factories.”
- “The Internet of Things (IoT) can be defined as a world of interconnected things that are capable of sensing, actuating, and communicating among themselves and with the environment (i.e., smart things or smart objects)”
- IoT as a network of networks
- The Internet of Things automates the user’s daily tasks, making them “smart” and capable of performing intervention actions while minimizing human interaction.
  - smart sockets can monitor power usage,
  - smart thermostats can provide better temperature control.
  - IoT sensors monitor equipment and alert business of maintenance needs, while smart logistics systems reduce fuel consumption.





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INTERNET OF THINGS (IOT) - TERMINOLOGY

IoT is not a single device, software or technology. IoT is a fusion of devices, networks, computing resources and software tools and stacks.

- **Things** - Every IoT device -- a thing or smart sensor -- is a small dedicated computer possessing an embedded processor, firmware and limited memory and network connectivity. The device collects specific physical data and sends that data out onto an IP network, such as the internet. Depending on the sensor's work, it might also include amplifiers, filters and converters. IoT devices are battery-powered and rely on wireless network connectivity through individual IP addresses. IoT devices can be configured individually or in groups.
- **Connections** - The data gathered by IoT devices must be transmitted and collected. This second layer of IoT involves the broad network, along with an interface between the network and back-end processing. The network is typically a conventional IP-based network, such as an Ethernet LAN and the public internet. Every IoT device receives a unique IP address and unique identifier.
- **Back end** - The large volume of real-time data produced by an IoT sensor fleet and collated at the IoT gateway must be analyzed to yield deeper insights, such as exposing business opportunities or driving machine learning.

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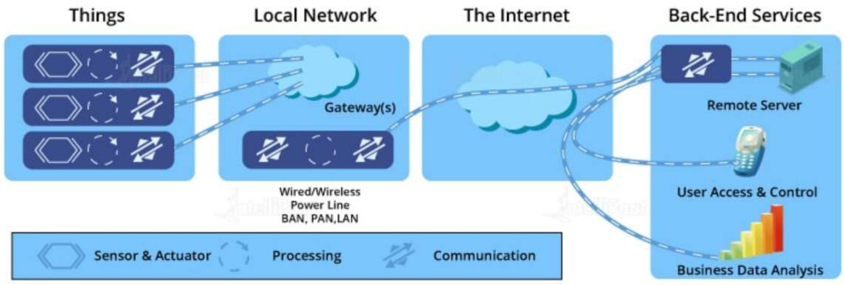
IoT SYSTEM COMPONENTS



- **IoT Devices**: smart electronic gadgets that consist of wireless sensors that help in transferring data over the Internet.
- **Local Network:** helps in accessing the data from the devices connected to the Internet.
- **Internet**: helps the devices connect with the user applications and servers.
- **Back-end Services:** They consist of a remote server, user access and control, and mobile applications.

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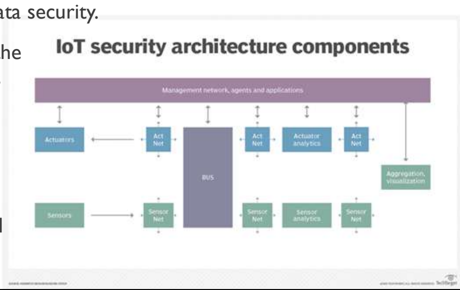
LAYERS OF AN IoT ARCHITECTURE

The discussion of sensor, connection and back-end layers can help business and IT staff understand IoT technology, The major architectural issues:

- **Infrastructure** - The physical layer includes IoT devices, the network and computing resources used to process the data. The infrastructure discussion often includes sensor types, quantities, locations, power, network interface and configuration and management tools.
- **Security.** The data produced by the internet of things can be sensitive and confidential. Passing such data across open networks can expose devices and data to snooping, theft and hacking. Organizations planning an IoT project must consider the best ways to secure IoT devices and data in flight and at rest. Encryption is a common approach for IoT data security.
- **Integration**. Integration is getting everything to work together seamlessly, ensuring the devices, infrastructure and tools added for IoT will interoperate with existing systems and applications -- such as systems management and ERP -- already in place in the organization.
- **Analytics and reporting**. The very top of an IoT architecture requires a detailed understanding of how IoT data will be analyzed and used. This is the application layer, which often includes the analytical tools, AI and ML modeling and training engines, and visualization or rendering tools.



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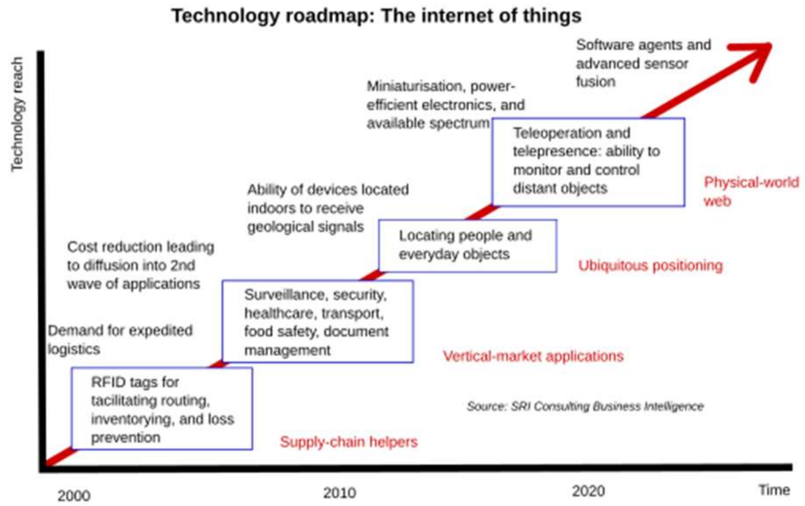
THE POTENTIAL OF IOT



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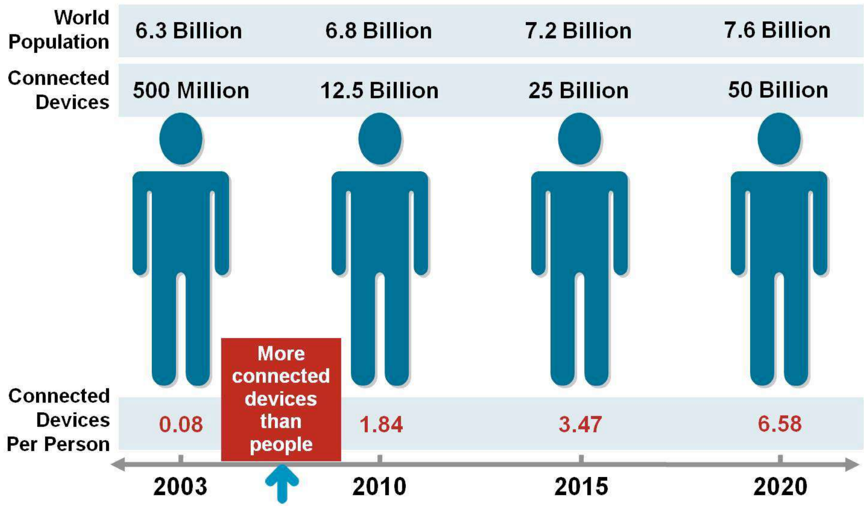
THE POTENTIAL OF IOT



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WHAT IS A "THING" ?

A *thing*, in the context of the Internet of Things, refers to any entity such as a device that forms a network and can transfer data with other devices over the network. In an IoT network, each thing has a unique identifier, might be part of an embedded system, and can collect and share data with minimal or no human intervention.

The role of "Things" in IoT

- IoT is composed of millions of things that are connected to each other, can share and upload data to the internet, and can share data with other things.
- Most things are embedded devices that can communicate with each other and the internet or an intranet.
- Some things, such as door locks, can perform their tasks mechanically.
- Others, such as washing machines, have built-in electrical circuitry.
- And some, such as smart cars, have computing capabilities. Some things in IoT are hardware devices while others are software products.

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IOT DEVICES

- **Smart home devices** - are interactive electronics that use wireless connections to understand user instructions. To an extent, smart home devices like thermostats and home security systems can work autonomously to assist with daily tasks.
- **Wearable technologies** - like Fitbits and Apple Watches connect to other devices (like your smartphone) to share data. They typically also connect to the internet to track GPS locations.
- **Personal medical devices** - like pacemakers are also IoT devices. Remote medical devices can help monitor and share a patient's vital signs or detect early signs of health issues for fast intervention.
- **Autonomous vehicles** - Self-driving cars and other connected vehicles rely on the internet to share real-time information. Sensors throughout the vehicle help map its surroundings, transmit camera footage, and respond to traffic signals.

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TYPES OF IOT APPLICATIONS

Billions of devices are connected to the internet, collecting and sharing information with one another. They range from smart home setups like cooking appliances and smoke detectors to military-grade surveillance equipment. The most common types of IoT applications:

- Consumer IoT refers to personal and wearable devices that connect to the internet. These devices are often referred to as smart devices.
- Industrial Internet of Things (IIoT) is the system of interconnected devices in the industrial sector. Manufacturing machinery and devices used for energy management are a part of the IIoT.
- Commercial IoT refers to the tools and systems used outside of the home. For example, businesses and health care organizations leverage commercial IoT for auditable data trails and consumer management.

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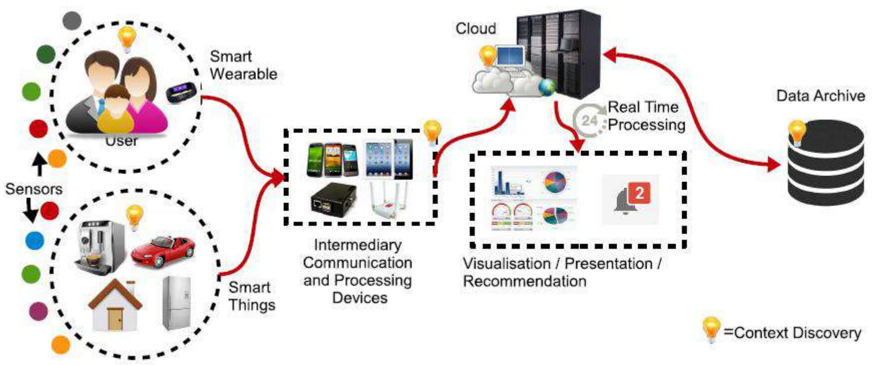
IOT ECOSYSTEM



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INTERNET OF THINGS COMPONENTS

- Internet of Things platform - An IoT platform manages device connectivity. It can be a software suite or a cloud service. The purpose of an IoT platform is to manage and monitor hardware, software, processing abilities, and application layers.
- Sensor technologies - IoT sensors, sometimes called smart sensors, convert real-world variables into data that devices can interpret and share. Many different types of sensors exist. For example, temperature sensors detect heat and convert temperature changes into data. Motion sensors detect movement by monitoring ultrasonic waves and triggering a desired action when those waves are interrupted.
- Unique identifiers - The core concept of the IoT is communication among devices and users. Unique identifiers (UIDs) establish the context of a device within the larger network to enable this communication. Identifiers are patterns, like numeric or alphanumeric strings. One example of a UID is the internet protocol (IP) address. They can identify a single device (instance identifier) or the class to which that device belongs (type identifier).
- Internet connectivity - Sensors can connect to cloud platforms and other devices through a host of network protocols for the internet. This enables communication between devices.
- Edge computing - It aims to conserve resources and speed up response time by moving computational resources like data storage closer to the data source. The IoT accomplishes this by utilizing edge devices like IoT gateways.
- Artificial intelligence (AI) and machine learning - Natural language processing (NLP) in IoT devices makes it easier for users to input information and interact with devices. One common example of an IoT device that utilizes NLP technology is the Amazon Alexa. Machine learning also enhances the analytical capabilities of IoT devices.

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BENEFITS/ DRAWBACKS OF IOT

Benefits of IoT	Drawbacks of IoT
Task Automation - Removing the need to perform daily tasks like turning the thermostat on and off or locking doors increases efficiency and quality of life.	Privacy - It can be challenging to protect the data mined by IoT devices. Increased tracking threatens the confidentiality of the information we share over the internet.
Conservation - Automation makes it easier to manage energy consumption and water usage without human oversight or error.	Security - Individual device security is left up to the manufacturers. Network security could become compromised if manufacturers do not prioritize security measures.
Analytics - Information that was previously difficult to collect and analyze can be tracked effortlessly with the internet of things.	Bandwidth - Too many connected devices on a shared network results in slow internet speeds.

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